

achieving lower cost requires defining the details of the application needs, so that specific processes can be developed and improved.

Silicon photonics has the potential to enable higher data rates with lower cost; however, multiple suppliers are developing competing solutions that are not interchangeable. Since data centers need multiple suppliers, it will be important to develop standards that enable pluggable replacement of components at some level. Even though different silicon photonic manufacturers use different manufacturing processes, the components need to operate inter-changeably.

4.2. LONG-TERM CHALLENGES

4.2.1. RF LONG TERM CHALLENGES

The biggest challenge for RF is to cost effectively increase the energy efficiency of >10GB/s communication for mobile devices. The energy efficiency of power amplifiers and DAC/ADCs decreases at frequencies above 10GHz. With 6G frequencies projected to be above 100GHz new technologies and circuits are needed to improve energy efficiency, but cost will need to be reduced for mobile devices.

4.2.2. OPTICAL INTERCONNECT LONG TERM CHALLENGES

In the longer term, standard interfaces will need to be developed for a number of high-performance optical interconnect technologies. To eliminate multiple optical-electrical-optical conversions in routing signals, optical information processing and logic needs to be developed. To increase optical processing density the third dimension needs to be utilized, as shown in Table OSC-3

The longer-term challenge that needs the most research is to enable optical routing functions. As mentioned earlier, significant time and energy is expended in converting optical signals to electrical signals in a router, decoding them to set the path and then converted to optical signals that are launched on the new path. Hybrid electrical/optical routing capabilities have been demonstrated in research to establish optical routing paths. More advanced capabilities with all optical routing may be capable of dramatically reducing the latency of communication between a CPU and memory or other CPUs.

Reducing latency in communication in data centers could enable applications of new architectures for communication between servers. Electro/optical routing is used in telecommunication and has been demonstrated with multiple technologies (i.e., MEMs) in “edge” applications. Thus, introduction of hybrid E/O routing in data centers may be efficient for sending large data streams.

If it were possible for optical logic to decode the routing instructions and change the optical path, this could significantly reduce the latency and potentially the energy of optical routing in a data center or local area network. Research is needed to identify materials, structures and devices that could perform optical logic such as decoding instructions, identifying paths that are available and switching the optical data stream to a new path.

Table OSC-3 Difficult Challenges

<i>Near-Term Challenges: 2024–2030</i>	<i>Description/ Summary of Issue(s)</i>
Achieving high frequency, energy efficient ADC/DAC technology compatible with CMOS processing.	• ADC efficiency degrades at sample rates above 50G samples/s (200Gb/s).
Components to meet 5G Performance Requirements.	• Increasing energy efficiency of amplifiers while increasing operating frequency. • Antennas to support multiple band communication in compact mobile devices. • High efficiency directional antennas are needed to increase range with low input power.
Integration of ASIC switches with optical I/O	• Establishing at least 2 suppliers with compatible optical I/O chiplets with high volume manufacturing.
Agreeing on standards for silicon photonic form factor and interfaces.	• Multiple manufacturers are developing competing solutions. Multiple manufacturers are developing competing solutions.

	• Technology development is expensive. • The market is small, so little incentive for competitors to develop standards.
5G mmWave Noise Cancellation.	• White noise is expected to be 30–36 dB higher than 4G, so circuits are needed to cancel the noise. • High density of RF signals are expected to interfere with mmWave communication, so circuits are needed to cancel these signals. • This may require use of noise phase resonators (e.g. photonic resonators) with multiple clocks.
<i>Long-Term Challenges: 2031–2039</i>	<i>Description</i>
Increasing the density of silicon photonics to reduce cost while reducing power consumption	• Optical devices are often linear or planar yet much could be done, especially to reduce size, utilizing the 3rd dimension. • Design, but specially fabrication in “Z” is “hard”. Some type of 3D printing might enable this technical solution. • Developing low power higher efficiency lasers and modulators requires new technology.
Develop methods for communication between systems with different wavelengths, polarizations, modulations.	• Technology is needed to up or down shift photons to different wavelengths. • Technology is needed to change polarization of light when entering a different system. • Devices are needed to translate and transmit photonic modulated information to a system with a different modulation scheme (i.e., Modulation Converters).
Reducing latency of communication between processors and memory or other processors in data centers.	• Signals from the processor must be serialized, converted to photons, and (photons converted to electrical be routed then converted back to photons multiple times). • Switches must arbitrate between conflicting routing request that adds latency. • Regenerate digital signals in the optical domain without returning to the electronic domain, a capability that is likely to require non-linear optical materials. • Perform logic operations in the optical domain. • Identify a strategy for reliable O/I with point-to-point connectivity. • May need a hybrid switch for small packet vs. large file messages.
Increasing energy efficiency of communication above 10 Gbps for 6G cost effectively.	• Identifying an energy efficient communication architecture and protocol. • Developing energy efficient components to support high data rate communication for 6G

5. TECHNOLOGY REQUIREMENTS

5.1. SUMMARY

In the IoE, communication from IoE device applications to the cloud is conducted with wireless, conventional copper, and optical interconnects. Within data centers, the cloud, communication is performed over copper and increasingly over fiber optical links and there is a need to increase data rates to support increasing traffic volumes. Increased use of artificial intelligence to learn on huge data sets is creating communication bottlenecks in data centers and high-performance computing centers, so new communication architectures and technology may need to be developed to mitigate these

*Table OSC-9 Satellites Capable of Connecting Standard Mobile Phones with the Internet***5.7.1. OFFICE AND FACTORY LOCAL AREA NETWORKS**

Offices and factories will have ever increasing need for bandwidth as more information is being transmitted by the internet of things devices and analyzed to improve business performance. Offices will require conventional services including messaging, virtual meetings, and access to data from the “cloud”, to analyze customer requirements, product availability, product performance (IoT) reports, etc. This will require increasing bandwidth over time that will require a combination of optical and wireless communications technologies, as shown in Figure OSC-1. In factories, information will be communicated between the tools in the manufacturing flow and manufacturing control where performance of individual tools will be monitored and issues “flagged”. This will require a local area network with connection to individual manufacturing tools, manufacturing control, technicians and engineers. The network should include a high bandwidth optical interconnect “back bone” connected to manufacturing control, fixed tools, and engineering analysis stations. Wireless connection could be used for communication with low data rate manufacturing tools and material handling systems that are mobile. The configuration of optical and wireless communication technologies will depend on the requirements of the office, factory, or warehouse. Optical interconnect technologies to be used could include high bandwidth fiber and/or a free space optical interconnect system (LiFi) with a wide range of wireless technologies including Wi-Fi, Bluetooth, LPWAN, custom wireless, and new wireless systems including 5G.

Medical facilities, such as hospitals, may have some tools or operating room areas that are sensitive to RF and EMI, so they may limit communication to shielded wire or optical interconnects for connectivity.

Since many technologies may be required to support these applications, only the optical interconnect LAN requirements will be discussed in detail in this section and others will be discussed in separate sections.

*Table OSC-10 Office and Factory LAN Requirements***5.7.1.1. OPTICAL LAN NEEDS**

The optical interconnect LAN requirement is projected to increase from current data rates of 100 Gb/s-wavelength with four wavelengths to 200 Gb/s-wavelength, as shown in Table OSC-10. Over this time, the power dissipation/Gbs will need to decrease by over a factor of four. To achieve these high data rates for each wavelength new technologies are required to compress more information in a single wavelength.

The biggest challenges for fiber optical LAN are to 1) increase data rates while reducing energy/bit and 2) reduce latency due to optical-electrical-optical conversion, as shown in Table OSC-5A.

5.7.1.2. OPTICAL LAN POTENTIAL SOLUTIONS

With silicon photonic modules becoming available, this should enable highly integrated power efficient communication. Furthermore, higher performance lasers that operate at lower power or highly efficient modulators that enable encoding more information in a single wavelength could enable evolutionary improvements in the silicon photonic modules. Furthermore, photodetectors and their amplifiers would need to operate at higher speed with lower energy per bit. Technology options to achieve higher data rates at lower power will be discussed in Section 6.

A second option for an optical interconnect LAN is “free space” optical interconnects (LiFi) that could operate using high data rate modulated LEDs[27] or lasers that are directionally controlled with mirrors or other means[28]. Several options exist for this technology that could use existing components, as shown in Table OSC-2; however, the data rates may be significantly below those in fiber as indicated in Table OSC-11. One advantage of an LED system is that it could easily communicate with machines that are moved frequently[27].

*Table OSC-11 Free Space Optical Communication - Key Attribute Needs***5.7.2. WIRELESS LANS**

Wireless will be used to communicate with computers, smart phones, smart appliances, tools, power management systems, and other IoT products, etc. Security will become an even more important issue to keep unwanted monitoring or malicious interference from outside wireless devices as more devices are monitored and controlled over the network. If Wi-Fi adopts

higher frequency for office LANs, e.g. by increased use of the 60 GHz band, which currently is the band used by 802.11ad and the upcoming 802.11ay(a) standard, this will reduce the leakage of data from buildings and make it more difficult to eavesdrop or hack the network. Furthermore, directional antennas could focus communication to the user and thus make it more difficult to eavesdrop on private communications. Furthermore, techniques could be developed that would detect signals coming from directions and distances other than the host network.

Small Business or Home

5.7.3. MOBILE SMART PHONES

As smart phones incorporate more functionality, they will need to communicate at higher data rates with the internet, but also detect signals from GPS satellites, cell towers (with mmWave or MIMO), non-terrestrial networks, health monitors, watches, and other RF sources, as shown in Table OSC-12. Thus, they will need to have compact antennas that can receive and transmit to multiple ranges of frequencies with multiple protocols. To communicate with mmWave and MIMO, they will need multielement antennas in arrays. The RF & AMS components will need to support all of the communication with high energy efficiency for multiple applications simultaneously.

A critical need for these devices is technology that enables highly secure operation and communication with the internet and other devices.

Also, with the increasing proliferation of wireless devices, there are significant concerns that interference between signals from the many wireless devices communicating in small areas will become a huge problem in the future. Some are proposing development of novel antenna and other schemes.

Table OSC-12 [Mobile Device Wireless Performance Requirements](#)

5.7.3.1. MOBILE SMART PHONE CHALLENGES

A significant challenge for mobile smart phones is to support the data rates required for new multimedia applications such as virtual reality while communicating with mmWave or MIMO. They will need to have multielement antennas that can effectively receive and transmit mmWave and MIMO at lower frequencies. While future smart phones need to communicate at higher data rates, they will need to operate with low power consumption. Furthermore, future smart phones will need to have improved security of personal information when communicating with the internet or other devices. An additional challenge is to integrate additional antennas into thin smart phones without causing interference in other communications or integrated circuits.

As applications require data rates higher than 10 Gb/s, the lower efficiency of the DAC/ADC and power amplifiers will make achieving communication efficiency numbers very difficult.

5.7.3.2. MOBILE SMART PHONE NEAR TERM POTENTIAL SOLUTIONS

In the near term, higher data rates would be supported by 5G technology that has a target of data rates 10 Gbps. The two options for this are multiple input multiple output (MIMO) at frequencies <6 GHz, or mmWave communication (28–78 GHz). Both of these technologies will require introduction of multiple antennas into the phone. Both are expected to employ directional communication from the base station to increase range while reducing interference and reducing transmitted power. A potential solution is the use of active phased array antennas to receive and transmit information. The use of directional antennas in both the base station and the cell phone may also allow implementation of methods to enhance security of communication. If 5G is initially implemented with <6 GHz, base stations would operate with massive MIMO and be able to support multiple users simultaneously, but the cell phone would operate with one or a few antennas. To improve communication throughput for the user, full duplex communication [29] is being considered to enable simultaneous transmission and reception by the cell phone and the base station. Full duplex would require new circuits to separate outgoing and incoming data.

To improve efficiency of ADC/DACs, new ADC concepts have been demonstrated that turn on converters only when processing data and operate with 6 bit accuracy and consumes 15.1 mW power consumption [30].

5.7.3.3. MOBILE SMART PHONE LONGER TERM POTENTIAL SOLUTIONS

In the longer term, high frequency mmWave may be used in high population density areas to support high data rate applications with a large number of users. A significant challenge is that 28 GHz to 78 GHz signals do not penetrate buildings, so buildings would need to have repeaters in the building or employ massive MIMO or ultra-high data rate Wi-Fi.

If 5G is implemented with mmWave and MIMO, cell phones would need to have multiple antennas to receive and transmit information and each antenna would need to have power amplifier (PA) and an ADC and DAC [31], which would consume considerable power operating with high precision at high frequencies. A critical challenge is to increase the energy efficiency of the PA at mmWave frequencies while maintaining linearity, since the efficiency of the PA decreases with operating frequency. Thus, higher performance of incumbent process technologies or even new materials may be required to improve energy efficiency, as shown in Figures OSC-6 and OSC-7. In the transmit mode, the antenna array would operate as an active phased array to focus transmission toward the base station. To overcome the high operating power of the high precision DAC/ADCs, use of hybrid analog/digital preprocessing [32, 33] or lower precision DAC/ADCs (1bit) [31, 32, 34] has been proposed. To compensate for potential blockages [35], multiple antennas would need to be integrated into the cell phone to be connected with multiple base stations.[36]

For >10 Gb/s communication rates, proposals have been made for a single bit zero crossing modulation protocol[new]; however, energy efficiency of all required components need to be investigated.

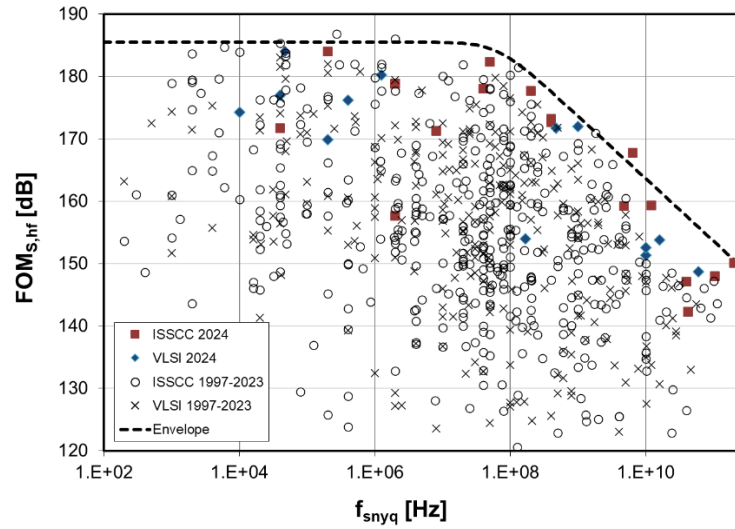


Figure OSC-6. ADC Performance Survey². $FOM_s = SNDR(dB) + 10 \times \log((f_s/2)/Power)$, where SNDR is measured at a high-frequency input near $f_s/2$.

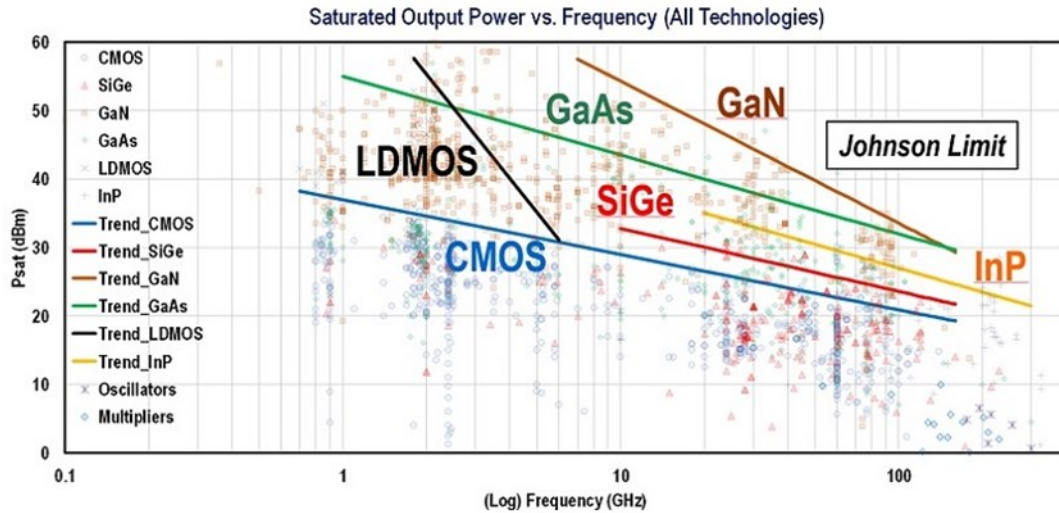


Figure OSC-7. Power Amplifier Output Power³

² <https://github.com/bmurmman/ADC-survey>

³ H. Wang, et. al. "Power Amplifiers Performance Survey 2000 to present." https://gems.ece.gatech.edu/PA_Survey.html.

5.8. AUTOMOBILES

Scope: Automobiles, trucks, freight trucks, heavy equipment, and farm equipment.

Autonomous vehicles will use wireless systems to communicate with networks, determine position and detect proximity to objects and vehicles. Because of the complex conditions and weather that autonomous vehicles need to operate, they will need a multitude of sensors to identify road signs, road hazards, pedestrians, animals, and classify the threats presented and anticipate their actions. They will also need to have access to traffic information and communication with other vehicles to identify changes in traffic flow that must be negotiated. The sensors could include video cameras, RF, microwave, “radar”, LiDAR, optical range detection, sonar and others. With the large number of different sensor types, it is proposed that a sensor fusion computer analyze their raw data to develop a map of obstacles. Currently, the forward-facing video cameras require the highest data rate for the 1080i of 1.5 Gbps for raw data, as shown in Table OSC-13. As automobile manufacturers seek to provide higher resolution images at longer distances, higher resolution cameras such as 4K will require data rates of >13 Gbps per camera for communication of raw data to the sensor fusion computer.

A significant challenge is that in heavy traffic, significant blind spots will be presented by vehicles in front of the target vehicle, so communication with other vehicles and local traffic management computers will be necessary to develop comprehensive maps of obstacles in the travel path will be required. Communication of information from sensors and guidance detectors will require significant amounts of data to be transmitted and analyzed, which will require use of high bandwidth optical or wired connections, as shown in Figure OSC-8.

Some have proposed that as many as six video cameras could be used by each vehicle. A significant unknown is how much information processing will be done at the sensor or camera prior to information being sent to the sensor fusion computer. It is assumed that data being received by vehicles from other vehicles or the cloud will be either processed images or maps of obstacles, so standards need to be developed for this communication.

Table OSC-13 Autonomous Vehicle Sensor Communication Data Rates

In the long run as autonomous vehicle guidance technology matures, several possibilities seem likely. Initially few of these vehicles will be operating so each will be a stand-alone source of data. As the number of vehicles on the road grows, benefits will arise from the vehicles communicating with one another. Possible benefits include vehicles sending object location to vehicles behind them, enabling fusing the location of both stationary (curbs, barriers, etc.) and moving (cars, trucks, bikers, pedestrians, etc.) objects so object locations are known as soon as possible. That information will enable vehicles to accommodate their motion sooner and with finer detail improving the object avoidance capability.

As the number of autonomous vehicles increases, another potential issue is interference between the optical and maybe even RF data streams. (Imagine the density and number of signals in the air on the 10 Freeway during rush hour in southern California where the freeway is more than six lanes wide in each direction with parallel on-off ramps and several levels of fly overs.) With LiDAR and Radar that broadcast electromagnetic wave beams, coupling of these will be needed to develop a dynamic 3D map of obstacles and moving hazards around the vehicle!

As the density of autonomous vehicles reaches a critical point, it is quite likely that vehicles will communicate sensor data to traffic control local (edge computers) instead of between vehicles. Then, the edge traffic control computer will broadcast an updated real-time map of obstacles to all vehicles within its radius of influence. In areas with lower traffic densities, the vehicle-to-vehicle communication will provide sensor data to vehicle sensor fusion systems that will provide the map of obstacles for the vehicle control system.

In the long term, (i.e., decades) when almost every vehicle, biker, pedestrian, etc. is visible electronically/digitally, one can see fusing motion data to enable merging of traffic streams at intersections, eliminating the need for traffic lights and enhancing capacity of the roads and highways and speeding traffic.

When vehicles become autonomous, they will need to be highly secured from hacking devices, and also monitor the health of the passengers to ensure that a medical emergency has not occurred, so sensors will be needed to monitor this and also potentially communicate with medical devices or smart phones. In early development of autonomous vehicles, significant amounts of data will be communicated to the manufacturer to assess actions to be taken and review decisions made by the processor in the vehicle. Security of control and passenger information will be paramount, so security systems must be able to quickly detect hacking or spoofed data that could cause danger to the vehicle and passengers.

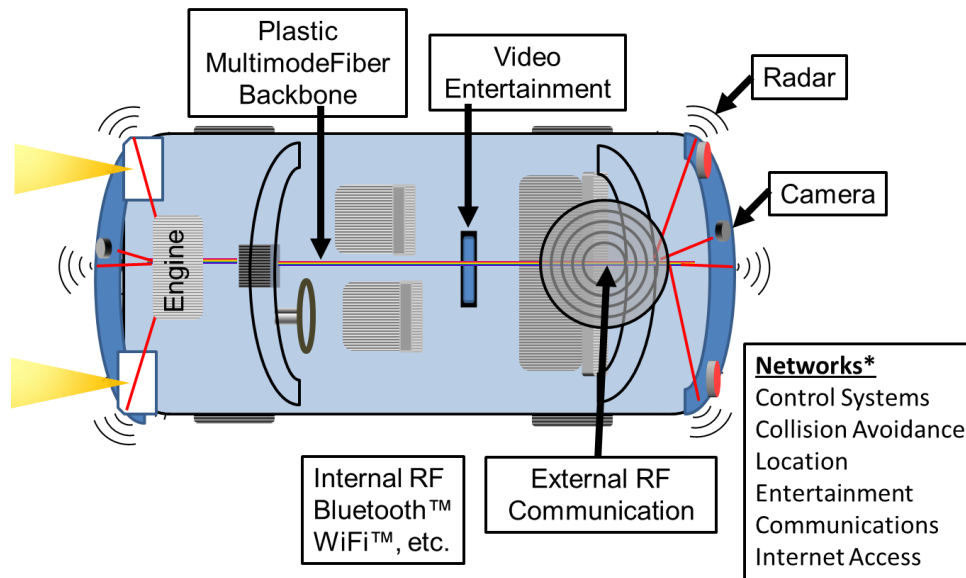


Figure OSC-8. Automobiles Have Multiple Networks to Measure Performance, Detect Obstacles, Control Operation, Deliver Entertainment to the Passengers, and Provide Communication to the “Cloud”

Note: As networks expand, there will be a move to reduce the use of copper and increase the use of low weight optical fiber interconnects with increased data rates and information density.

5.8.1. WIRELESS COMMUNICATION CHALLENGES

As vehicles strive to become more autonomous, more wireless capabilities will be employed to communicate with Edge computing networks through terrestrial and non-terrestrial networks (NTN), traffic networks to detect and monitor vehicles, pedestrians, animals, bicycles, obstructions, highway construction, and weather-related hazards, etc. This would include GPS and radar-like detectors to monitor proximity, trajectory and speed of vehicles, pedestrians, animals, etc. For communication through NTN, the vehicle will need to communicate with moving satellites while the vehicle is moving, so needs to seamlessly connect to nearer satellites and account for Doppler effects. Also, this information would be coupled with video and image recognition to determine expected actions of the objects. For the radar-like devices, different frequencies may be used to determine the density of objects. When the vehicle is traveling in inclement weather (e.g., fog, rain, snow, wind, flooding, etc.), multiple sensors will need to communicate with the central processor and internet, and protocols established on which monitors should be most “believed” in specific weather conditions. As auto vehicles become more autonomous, the demand for high-speed communication, data, and entertainment media will increase. Within the vehicle, a Bluetooth and/or Wi-Fi like network will be used to communicate with portable devices including computers, phones, watches and possibly health monitors.

5.8.2. WIRELESS COMMUNICATION POTENTIAL SOLUTIONS

5G data rates with either massive MIMO or mmWave could support high data rate communication of maps, traffic flows and data from the vehicle to the manufacturer. Massive MIMO at frequencies <6 GHz has the potential for longer coverage range by a base station, while mmWave would have a shorter range of communication. Both approaches could use directional antennas with tracking of base station location to reduce the risk of spoofing and eavesdropping. This would require active phased array antennas with circuits and software to analyze incoming signals.

5.8.3. OPTICAL NETWORKS

With automobiles having more electronics and sensors for more autonomous driving vehicles, there is a need for high data rate and low weight communication networks. As a result, some automobiles are using plastic optical fiber communication to reduce weight and increase data rates. Over time, the data volumes are expected to increase as more sensors including cameras, radar, environmental sensors, and engine performance sensors are added to identify obstacles and improve automobile performance and comfort. To support this, data rates in fibers will increase from current 40 Gb/s to 100 Gb/s in 2023, as shown in Table OSC-14. A significant concern with plastic optical fiber is the practices and environment used for vehicle assembly and repair. Some commercial ICs are being developed to communicate over copper wire with the required data rates, so it is not clear which solution will prevail. It is assumed that current and developing technologies will be able to support these data rates in the required timeframe.

Free space optical may be used to provide information on proximity to other vehicles through lighting system modulation. In fact, the modulation of the brake lights could inform the vehicles behind whether it was just a “light touch” on the brake pedal, or if it were a hard break due to a dangerous situation. This was already implemented years ago by having different intensities for the brake lights, but that may not always work very well, depending on the environment conditions (fog, rain, snow, etc.) and distance between vehicles.

The biggest challenges will be to meet the environmental requirements for different applications that will require robust packages to manage shock and vibration, temperature extremes, etc. In addition, the fiber optic networks will need to meet significant cost challenges, which will require increasing levels of integration at the package and then at the device level. These increasing levels of integration can be supported by technologies developed to support Office LAN and data centers.

Table OSC-14 Automotive Optical Interconnect Requirement and Potential Solutions

5.8.3.1. OPTICAL NETWORKS POTENTIAL SOLUTIONS

Plastic optical fiber is the medium of choice and currently operates at data rates of 150 Mb/s for entertainment and is expected to increase to 1 Gb/s in 2020 and 10 Gb/s by 2025. For sensor and control communication in autonomous vehicles, data rates of 40 Gb/s are expected to increase to 100 Gb/s in 2023. To support higher data rates in the future, silicon photonic modules could be used with the plastic fiber; however, they would need to operate over a wider temperature range and at wavelengths compatible with the plastic fibers.

5.9. AEROSPACE

Fiber optics and wireless communication are being used in aircraft applications to reduce weight and increase bandwidth for a range of applications. Applications include entertainment, in flight networking, display systems and RF over fiber communication. In-flight entertainment can be transmitted over fiber to a “box” near the seats. For in-flight networking, data can be transmitted over fiber or free space optical to Wi-Fi stations or seat consoles that communicate with customer portable or mobile devices, as shown in Figure OSC-9. Furthermore, data from sensors, and potentially from control systems, can be communicated over fiber optics to the pilots. Fiber optics communication has several advantages over conventional copper wires including lower weight, electromagnetic interference immunity and smaller size.

High flying drone-like aircraft are also being developed to provide internet access with RF over large areas, as shown in Figure OSC-10, while the circling over an area needing access. This capability could be quite valuable to provide internet and mobile phone support for areas struck by disasters, earthquake, typhoons, hurricanes, floods, etc.

Free space optical interconnects are being proposed to provide communication, internet access, and entertainment to passengers in aircraft to reduce cost and avoid installing fiber in barriers [37, 38]. Free space optical interconnects are also proposed to support internal avionics communication [39] because the signals are secure from outside detection. These interconnects could use white LEDs, lasers, etc. and are immune from EMI, are low weight, low cost, and low power.

Currently, fiber optic communication is available in aircraft to operate at 10Gb/s, in multi-mode or single mode, and with wavelengths of 850 nm, 1310 nm, and 1550 nm. Table OSC-15 Fiber optic devices in aircraft must be able to operate in harsh environments, with thermal shock, mechanical shock, extreme temperatures (-40 to 125 °C), and devices must be protected from lightning strikes. It is anticipated that more applications will emerge related to increased use in control systems.

Table OSC-15 Aerospace Optical Interconnect Requirements (Preliminary)

*Table OSC-16 IoT Wireless Performance Requirements (Battery)***5.11.1. IOT POTENTIAL SOLUTIONS**

For the battery powered IoT devices, the potential solution is to develop more energy efficient communication protocols. For security, the IoT hub in addition to encryption may need to detect the location of the transmitting IoT device and verify that it is authorized. 5G also plans to support lower frequency (<1 GHz) communication with low data rate-low power devices over the same radio as higher data rate applications.

For powered high data rate applications, 5G communication with massive MIMO or mmWave would provide a data rate of 1 Gb/s. Furthermore, the directional communication used with massive MIMO and mmWave would reduce the likelihood of eavesdropping or spoofing. This could be further improved by including circuitry that continuously tracks the location of the IoT device and “hub” and detects anomalies.

6. INTEGRATED CIRCUIT AND DEVICES

Higher data rate lower power electronic integrated circuits and devices are needed to support both higher data rate RF communication and optical interconnects. For higher data rate RF employing pulse amplitude modulation (PAM) and other modulation techniques, lower power digital to analog converters (DAC), higher efficiency power amplifiers with high linearity and higher sample rate ADCs are needed for base stations and mobile devices. Furthermore, highly directional phase array antennas are needed to focus RF signals to the user instead of broadcasting in all directions. Optical interconnects also need higher data rate lower power electronic driver circuits and receiver circuits to support higher data rate optical transmission and receivers. Optical interconnects also are employing optical integrated circuits to increase data rates and potentially reduce power consumption. To continue increasing density of integration and reducing power per bit, lower power higher efficiency more compact optical sources and modulators are needed. In addition, more compact waveguides and multiplexers are needed to increase density without increasing crosstalk to reduce cost per bit communication. Optical receivers need high speed, highly efficient, high linearity compact optical detectors and amplifiers to support PAM and other modulation of signals in the fiber. For both photonics and high-speed RF, terahertz electronics may be needed to provide driver circuits that provide well shaped signals for precision required to support high levels of modulation.

6.1. RF INTEGRATED CIRCUITS AND DEVICES

The challenges for RF components include 1) achieving high performance, wider bandwidth to increase data rate, and energy efficient RF analog technology with high linearity that is compatible with CMOS based digital processing that has a good compromise for high data rate (wide band) ADC/DAC, and 2) components to meet 5G and 6G performance requirements. While all circuits contribute to loss of energy efficiency, the energy efficiency of PAs based on a given FET technology decreases with increasing mmWave frequencies; linearity can also be degraded.

6.1.1. INTEGRATED CIRCUITS FOR RF OR OPTICAL INTERCONNECTS

To increase data rates of communication with lower power and lower power, DAC and ADCs need to operate with higher sampling rates and lower power consumption. These optical interconnections have been developed to support huge data transmission. Moreover, to improve the whole link capacity further, recently, a few companies have developed integrated silicon photonic “chips” that are “self-aligned” to fibers with advanced multiplexing technologies, including wavelength-division multiplexing (WDM) [41], mode-division multiplexing (MDM) [42], and polarization-division multiplexing (PDM) [43]. Also, serializer-deserializer (SERDES) circuits need to operate at higher frequencies with higher efficiency. To achieve higher data rates with improved energy efficiency in these circuits, devices need to operate with lower power at higher frequencies. This may require introduction of new device, gate, contact, or interconnect materials while decreasing circuit cost.

Recently, a 5 nm integrated circuit technology was announced that demonstrated a transmitter test circuit that operated at 130 Gb/s data rates with PAM4 modulation and 0.97 pJ/bit energy consumption [44]. For ultra-high-speed data converters in 5 nm technology, the state-of-the-art is defined by receiver chip described in [45], featuring a 200 GS/s ADC/DAC combo. The ADC maintains approximately 6 effective bits for high-frequency inputs and dissipates only 400 mW, leading to outstanding figures of merit (Schreier FOM of 153 dB, Walden FOM of 38 fJ/conv.-step.). Generally, the FOM improvement in ADCs has been relatively slow, as illustrated in the figure below. Low-frequency designs are currently approaching practical limits in power dissipation as imposed by thermal noise (the red line in the figure corresponds to the well-known $8kT \times \text{SNR}$ energy limit, corresponding to $\text{FOM}_S=192$ dB) and may not improve much further. In contrast, high-

frequency, low- to moderate-resolution designs will continue to benefit from device technology improvements that enable higher levels of parallelism and lower dynamic power (CV^2f).

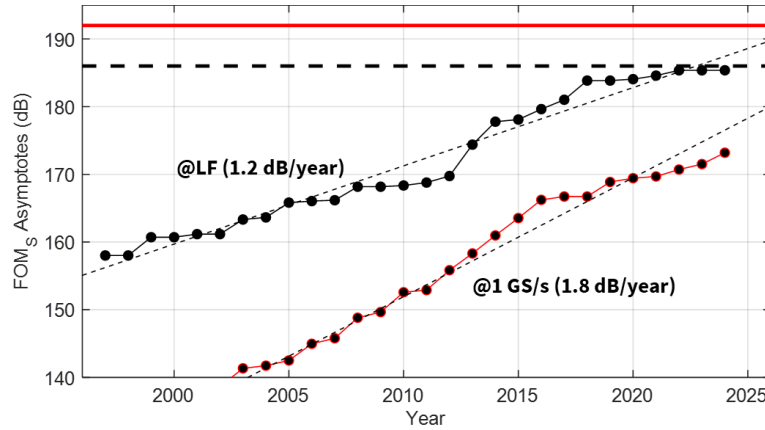


Figure OSC-12. ADC FOMs asymptotes over time (see Figure OSC-6 for a plot of the raw data). Low frequency ADCs tend to improve by 1.2 dB/year, whereas high-speed designs improve by 1.8 dB/year. Source: B. Murmann.

6.1.2. CMOS DEVICES

The 2024 CMOS Logic roadmap indicates the technology transitions from a FinFET with L_g of 16nm in 2024 to a stacked gate all around structure (GAA) in 2025 with an L_g of 14nm. Beyond this, L_g is gradually reduced to 10nm by 2033 and remains constant through 2039. The RF-AMS performance predictions for CMOS devices have been restricted to the metrics peak f_T (Figure OSC-12) and peak g_m (Figure OSC-13). The 2024 roadmap gives the above performance FoMs for n-channel FDSOI and double-gate FinFET high-performance devices obtained from technology computer aided design (TCAD)-based device simulation [46] that has been calibrated on recent averaged measured data for the 28 nm, 22 nm, and 16 nm nodes. These simulations were based on hydrodynamic transport with thin silicon mobility physics, as well as the estimated resistive and capacitive device parasitics up to the first metal layer, which defines the terminals of a transistor cell in high-frequency analog circuit design. Similar simulation of surround gate structures have not been completed, however, restricting the L_g to 10nm in surround gates may avoid performance degradations due to surface scattering below 10nm. The FinFET simulations indicate the degradation in f_T and g_m at gate lengths below 10 nm are a result of mobility degradation caused by surface scattering at the gate oxide interface, and due to the ever-thinner silicon body. As can be observed, the double gate of the FinFET results in higher transconductance but also higher capacitive parasitics compared to the single-gate FDSOI MOSFETs. Other MOSFET high-frequency figures of merit, such as f_{MAX} and NF_{MIN} , which are strongly dependent on the device layout used in particular circuit blocks have been removed from the CMOS section of the OSC tables. Note that the displayed f_T (Figure OSC-12) does not include the capacitive parasitics resulting from connecting the transistor cell to the passive devices in the upper metal layers of the back-end. The introduction of new materials, with stacked surround gate and gate structures may change the f_T trend of 14nm to 10nm CMOS devices. Reduction of gate resistance has in the past been demonstrated to improve f_{MAX} [47, 48] and noise figure, while reducing the source/drain contact resistance improves both f_{MAX} and f_T .

Table OSC-18 Group III-V Compound Semiconductor FET and Bipolar Transistors Technology Requirements

The unique challenges for III-V devices are yield (manufacturability), substrate size, thermal management, integration density, dielectric loading, and reliability under high fields. Challenges common with Si-based circuits include improving efficiency and linearity/dynamic range, particularly for power amplifiers. A major challenge is increasing the functionality of power amplifiers in terms of operating frequency and modulation schemes while simultaneously meeting increasingly stringent linearity and efficiency requirements at the same or lower cost.

6.1.4.1. GROUP III-V COMPOUND SEMICONDUCTORS CONSISTING OF ELEMENTS FROM GROUPS III AND V [BOTH BIPOLAR AND FIELD EFFECT TRANSISTORS (FET)]

For Group III-V devices, potential solutions under evaluation include improved thermal management, new epitaxy and substrates, device scaling technologies, heterogeneous integration on silicon, high throughput e-beam lithography (FETs only), and multilevel interconnects, as shown in Figure OSC-23. More detailed descriptions of these can be found in [51, 52,53].

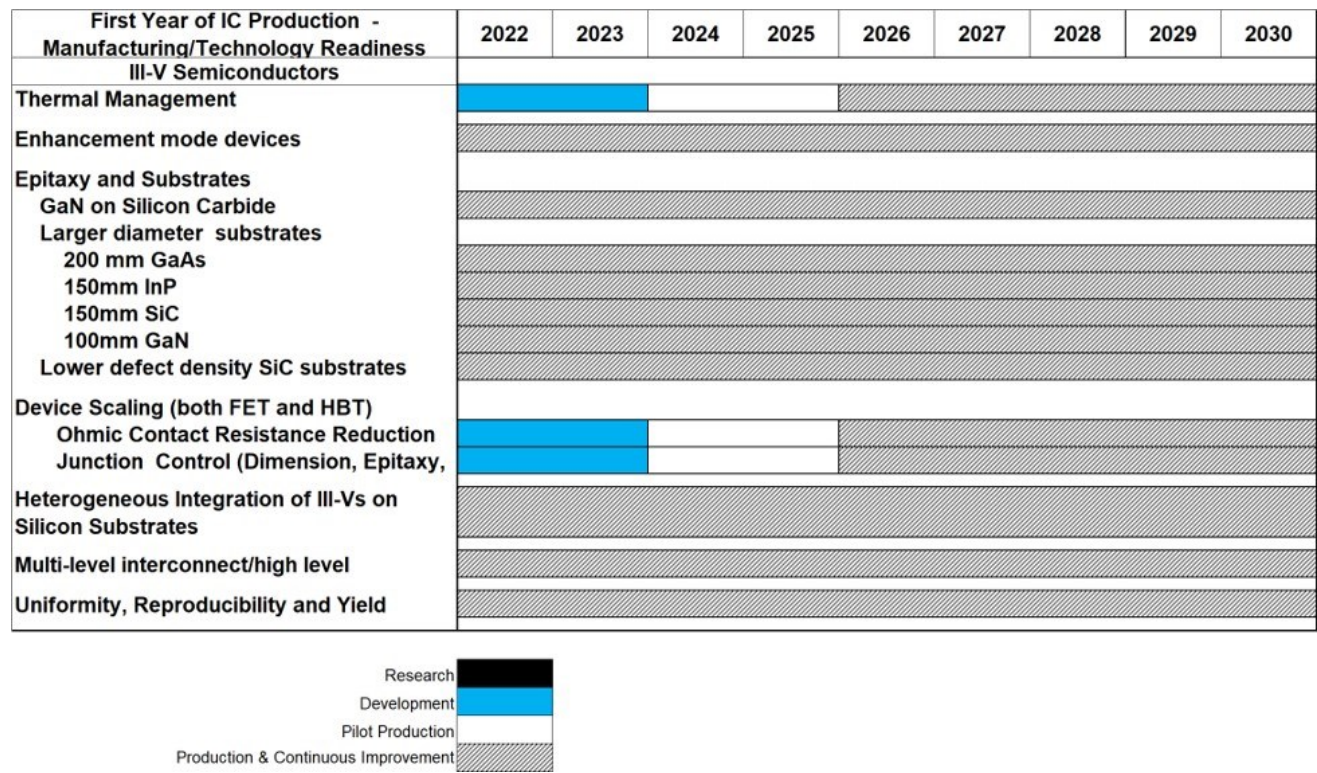


Figure OSC-23. *Group III-V Compound Semiconductors Potential Solutions*

6.1.5. ANTENNAS TO SUPPORT 5G+ COMMUNICATION

Both mMIMO and mmWave based communications are proposing to use highly directionally focused beams, to and from the user, to increase range while using less power.

The potential solution is to employ phased array antennas where signals of multiple small antennas are shifted to constructively reinforce in the desired direction and destructively interfere in other directions. Several groups have demonstrated these on silicon substrates while others are fabricating them on laminate substrates. The challenge is to have energy and component efficient operation to meet cost and energy objectives. With phased array antennas, the signals must be timed to constructively interfere and change direction as the communication direction changes. Base stations may employ more sophisticated control, but mobile smart phones need simple, low cost, low power techniques.

For mobile phones, the cell phone would need to have multiple antennas to receive and transmit information and each antenna would need to have a PA and an ADC and DAC [31], which would consume considerable power operating with high precision at high frequencies. In the transmit mode, the antenna array would operate as an active phased array to focus

transmission toward the base station. To overcome the high operating power of the high precision DAC/ADCs, use of hybrid analog/digital preprocessing [32, 33] or lower precision DAC/ADCs (1bit)[28, 30, 31] has been proposed. To compensate for potential blockages[35], multiple antennas would need to be integrated into the cell phone to be connected with multiple base stations.[36]

Since the mmWave focusing antenna has multiple elements transmitting signals with different phases, the antenna array would need to be fabricated on a smooth substrate with low k dielectric and precise control of feature size shape, and separation. For 28 GHz the $\frac{1}{4}$ wavelength is ~ 0.25 cm and at 72 GHz the $\frac{1}{4}$ wavelength is ~ 0.1 cm, so distortions of some fraction of this could cause problems with beam forming. Substrates with variation in surface height could result in phases being shifted resulting in changes in the direction of the focused beam. Similarly, the phased array substrate must not warp with temperature or the beam focus and direction could shift with operating temperature. Thus, thin substrates with high modulus and low dielectric constant will be needed as frequencies increase to low mm wave range.

Since the number of antennas in a mobile phone continues to increase, a switchable antenna has been proposed that could switch between diversity mode and MIMO mode [54]. Also, an antenna has been proposed that could receive 4G and 5G (3.5 GHz MIMO) [55].

6.1.6. TERAHERTZ ELECTRONICS

Radiation in the terahertz (THz) range, which is non-ionizing and enables imaging through opaque material layers, is of potential relevance for a large variety of applications [56, 57] such as: non-destructive material inspection, medical diagnostics, transportation, security screening, industrial automation, extremely high data-rate communications, and space exploration. Unfortunately, THz signal power generation at room temperature (RT) by either electronic or optical components presently suffers from poor device performance. Particularly, between 0.5 and 10 THz the output power of both optics and electronics drops to very low levels. Similarly, signal detection in this THz gap is plagued by high loss in the frequency conversion process. The realization and deployment of future, partially mobile, RT-integrated THz systems will be driven mainly by cost, form-factor, signal power, and energy-efficiency. These requirements and the much easier integration with digital signal processing circuitry favor electronics-based systems over purely optical systems.

Semiconductor technologies with transistor cut-off frequencies far beyond 300 GHz (the lower limit of the so-called THz gap) have been surveyed in [49, 51, 52, 57, 58]. The performance of various circuit blocks based on present and future (expected) HBT technology have also been assessed [52 and 59]. One of the results of the European DOTSEVEN project was the first all-silicon computer tomograph operating in the THz gap using SiGe HBTs for radiation and Si CMOS based sensors. Monolithic ICs have also been demonstrated with InP High Electron Mobility Transistors (HEMT) [60,61].

To enable precise shaping and decoding of modulated signals, THz electronics may be needed to provide the drivers, amplifiers and detectors for both photonics and RF. With 5G pursuing both <6 GHz massive MIMO and mmWave RF, technologies beyond 5G may require 100GHz+ operating frequencies. Furthermore, photonics is increasing data rates in single fibers through a use of multiple wavelengths and modulation of different photonic properties. In either case, electronics is needed to process the data stream and feed data to the CPUs. THz electronics will be needed in the future to support these high communication data rates.

A major challenge of THz electronics is the loss of output power with frequency. Thus, obtaining gain and product-level output power will require transistor cut-off frequencies well beyond 1 THz. Other challenges are the development of suitable low-cost THz packages as well as measurement techniques and equipment for standard device and circuit characterization. Therefore, opening up the THz gap for electronic applications benefiting society will be require innovations at the device, material, and system architecture level as well as in circuit design techniques and modeling.

6.2. PHOTONIC INTEGRATED CIRCUITS AND DEVICES

Near-term challenges are to 1) Increase the operating frequency and density of optical transceivers while reducing their power and cost, and 2) develop router/switch package integrated photonic interconnects for data centers. Long-term challenges include 1) processing information in the optical domain, and 2) develop methods for communication between systems with different wavelengths, polarizations, modulations. [62]

6.2.1. PHOTONIC MONOLITHIC HETEROGENEOUS INTEGRATED CIRCUITS(PIC)

Recently, optical monolithic heterogeneous integrated circuits have been introduced into the market by multiple suppliers. Currently, most of these circuits have separate transmitter and receiver circuits.

Some of these transmitter circuits have the lasers integrated with the modulators, while some of these have the laser packaged separately from the modulators. In most cases, the modulators and waveguides are single crystal silicon on